DGMS 2019 • SECTION K: ENGINEERING GEOLOGY & DAM STRUCTURES

Question 83: Dam Structural Types: Arch Dam Load Transmission to Abutments

Official Examination Guide • Structural Mechanics, Arch Action vs. Gravity Transfer & Abutment Geology

OFFICIAL EXAMINATION QUESTION

In which one of the following dams, the major portion of the water pressure is transmitted to the abutments or sides of the valley?

(a) Earth dam	**(b) Arch dam** CORRECT ANSWER
(c) Buttress dam	(d) Rock-fill dam

1. Executive Summary & Core Principle

Dams are classified into structural types primarily by how they resist and transmit the immense hydrostatic horizontal water pressure of the impounded reservoir:

- **Arch Dam:** An arch dam is a solid concrete or masonry structure curved upstream in plan (convex face towards the water). Through **arch action (ring compression)**, the horizontal hydrostatic load is resolved into tangential thrust forces that are transmitted laterally into the canyon walls or rock **abutments (sides of the valley)**! Because of this horizontal load transfer, the dam relies relatively little on self-weight, allowing for a much thinner cross-section than a conventional gravity dam.
- **Earth Dam & Rock-fill Dam (Embankment Dams):** Massive trapezoidal embankments composed of compacted soil, clay core, and rock fragments.

They resist water pressure entirely by dead weight and internal shear strength, transferring the hydrostatic thrust directly downwards across a broad footprint into the **foundation bedrock and riverbed**.

- **Buttress Dam:** Consists of a continuous upstream deck (flat slab or multiple arches) supported on the downstream side by a series of equidistant triangular reinforced concrete walls (buttresses). The loads are transferred down the buttress walls into the **foundation bedrock**.

Therefore, the dam type in which the major portion of water pressure is transmitted to the abutments or sides of the valley is the **(b) Arch dam**.

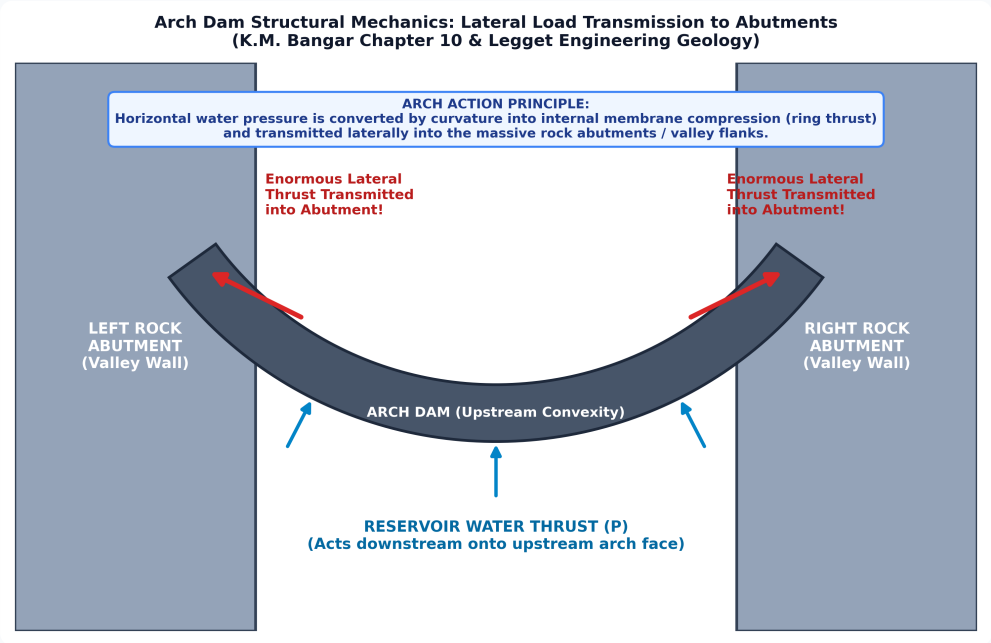


Figure 83.1: Plan-view schematic illustrating load transmission mechanics in an Arch Dam: upstream water thrust is converted via structural curvature into lateral membrane compressive thrusts driving directly into the massive valley rock abutments (Grounded in Bangar Engineering Geology and USBR Design Standards).

2. Comparative Load-Transfer Mechanics of Major Dam Types

Dam Type	Primary Resistance Mechanism	Load Transmission Path	Ideal Topographic & Geological Setting
Arch Dam	**Arch action** / **Ring compression** (structural shape curvature)	**Laterally transmitted into the valley sides** / **rock abutments**	**Narrow, deep V-shaped canyons or gorges** with exceptionally strong, massive, unweathered rock flanks (granite, gneiss, basalt).
Gravity Dam	Self-weight of concrete/masonry resisting sliding and overturning	Vertically and obliquely downwards into the **foundation floor**	Narrow to moderate valleys with sound, competent bedrock across the entire river floor.
Buttress Dam	Upstream water weight pressing onto inclined deck, stabilized by buttresses	Transmitted down triangular buttress footings into **foundation**	Wide river valleys where strong foundation rock exists but concrete economy is desired.
Earth Dam	Massive self-weight and internal friction/ cohesion of compacted earth	Transmitted over broad base footprint into **foundation soil**/ **rock**	Broad, flat river plains or valleys with weak, permeable, or alluvial foundations.
Rock-fill Dam	Self-weight of loosely packed/ compacted rockfill with impervious membrane	Transmitted downward over wide base into **foundation bedrock**	Wide valleys with readily available quarry rock; tolerates slight differential settlement.

3. Geological Requirements for Arch Dam Abutments

Because an arch dam transmits tens of thousands of tonnes of lateral compressive thrust into the valley flanks, the geological integrity of the **abutments** is far more critical than in any other dam type:

1. **High Compressive Strength:** Abutment rocks must possess high uniaxial compressive strength (UCS $> 100\text{ MPa}$) to resist crushing under concentrated arch thrusts.
2. **Absence of Downstream-Dipping Joints:** Joint sets or foliation planes that strike parallel to the river and dip downstream into the canyon create severe wedge-sliding failure hazards behind the abutment thrust blocks.
3. **Modulus of Elasticity Homogeneity:** Both the left and right abutments must have comparable rock mass stiffness (E_m) to prevent asymmetrical deflection and differential twisting of the arch crown.
4. **Famous Indian Landmark:** The **Idukki Dam** in Kerala (168.9 m high) is India's premier double-curvature parabolic arch dam, constructed across the narrow granitic gorge between the Kuravan and Kurathi hills.

4. Systematic Option Evaluation

- **(a) Earth dam: INCORRECT.** Transmits load vertically downward into the foundation base over a very wide footprint.
- **(b) Arch dam: CORRECT ANSWER.** By virtue of its upstream curvature and arch action, hydrostatic water pressure is converted into compressive ring thrust and transmitted directly to the valley abutments.
- **(c) Buttress dam: INCORRECT.** Transmits deck load down vertical triangular buttresses into the foundation bedrock.
- **(d) Rock-fill dam: INCORRECT.** An embankment dam transmitting forces vertically downwards into the valley floor.

> **Key Takeaway:** Arch Dam = Transmits major water pressure laterally to valley abutments; Gravity/Earth/Rock-fill Dams = Transmit load to base foundation. Hence, **(b) Arch dam** is the correct answer.

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